PRESENTATION

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AGENDA

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INTRODUCTION

- **A machine learning project that aims to help factories predict machine failures before they occur by analyzing sensor data and building a model that predicts the probability of a machine failure within the next 7 days.**

1

Target

Predict machine failure within the next 7 days (Failure_Within_7_Days).

2

Data

500,000 records of factory machine sensor data (Factory Sensor Data).

3

Output

A trained and deployment-ready classification model (model.pkl).

PROBLEM OVERVIEW

- **Production downtime & financial losses**
- **Reactive/Preventive maintenance**
- **Predictive maintenance with sensor data**
- **Imbalanced dataset**

~6%

Only a small percentage of machines represent actual failure cases (Failure = True) in the dataset.

Class Imbalance

CURRENT SOLUTIONS

Reactive Maintenance	Preventive Maintenance	Predictive Maintenance
Repair after the failure occurs	Scheduled maintenance at fixed intervals.	Predict failures using machine learning and sensor data before they occur.
✗ Unexpected downtime and high financial losses.	✗ Higher costs and sometimes unnecessary maintenance.	✓ Minimizes downtime and reduces maintenance costs.

DATASET & EDA

factory_sensor.csv

Rows	**500,000**
Columns	**22**
Target	**Failure_Within_7_Days**

Key Data Preprocessing Steps

- Removed the Machine_ID column.
- Applied One-Hot Encoding to the Machine_Type column.
- Filled missing values with 0 using fillna().

DATA PREPROCESSING – CODE

```
df.drop(columns=['Machine_ID'], axis=1, inplace=True)

df = pd.get_dummies(df, columns=['Machine_Type'], dtype=int)
df.fillna(0, inplace=True)

# leaky columns
df.drop(columns=['Remaining_Useful_Life_days',
                  'Operational_Hours'], axis=1, inplace=True)

# low-correlation columns
df.drop(columns=['AI_Override_Events',
                  'AI_Supervision'], axis=1, inplace=True)
```

DATA PREPROCESSING

1. Removed non-essential columns (Machine_ID).
2. Applied One-Hot Encoding to the Machine_Type column.
3. Filled missing values (fillna)).
4. Removed data leakage (leaky) columns.
5. Removed features with low correlation to the target.
6. Converted all values to float.

WORKFLOW

Raw Data
EDA
Cleaning & Encoding
Feature Selection
Train/Test Split
SMOTE Oversampling
Model Training
Evaluation
Save Model (joblib)

MODELS

Random Forest

- n_estimators = 100
- max_depth = 15
- n_jobs = -1
- random_state = 42

XGBoost

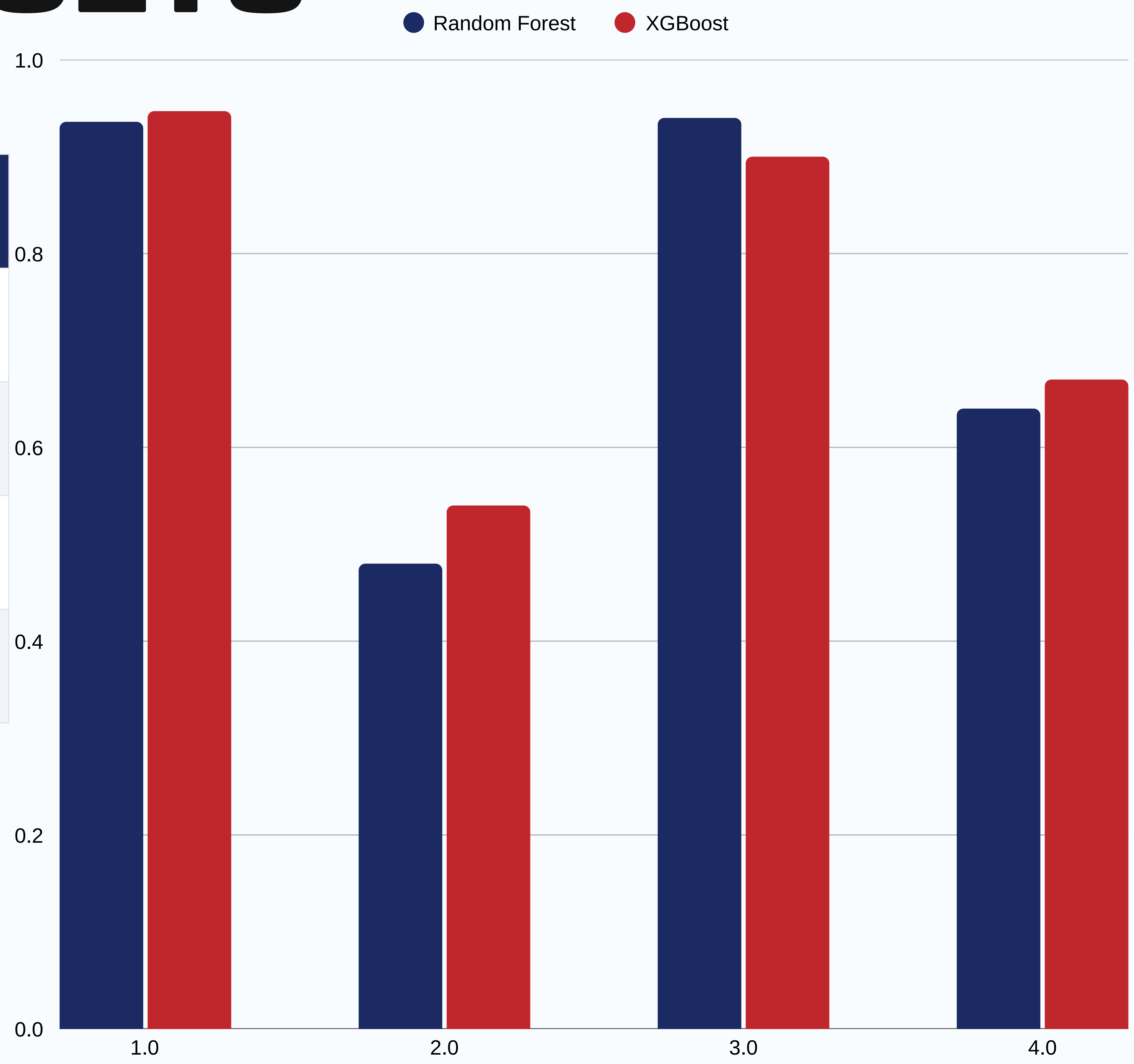
- n_estimators = 200
- learning_rate = 0.05
- max_depth = 5

RESULTS

Test Set Performance

Metric	Random Forest	XGBoost
Accuracy	93.6%	94.7%
Precision (Failure)	0.48	0.54
Recall (Failure)	0.94	0.90
F1-score (Failure)	0.64	0.67

- **XGBoost achieved the highest overall performance, but both models still require improvement in precision to reduce false alarms.**



CHALLENGES

! Missing data

Four sensor columns contained more than 90% missing values.

! Class imbalance

Actual failure cases account for less than 6% of the dataset.

! Low precision

Relatively high false alarm rate for the failure class.

! Data leakage columns

Some features leaked information directly about the target variable.

FUTURE WORK

1. Improve precision to reduce false alarms.
2. Use time-series techniques (e.g., LSTM) to capture machine behavior over time.
3. Apply additional feature engineering to sensor data.
4. Optimize hyperparameters using Grid Search or Bayesian Optimization.
5. Build a dashboard to display predictions in real time.

THANK YOU